

Classical Forecasting Techniques

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Classical forecasting techniques are a set of methods used to predict future values based on historical data. These techniques are widely used in various fields, including finance, economics, and supply chain management. In this section, we will explore some of the most commonly used classical forecasting techniques, including naive methods, moving average methods, exponential smoothing methods, Holt method, and Holt-Winters methods. We will also discuss their advantages and limitations, as well as how to implement them in practice.

1 Data Generating Process (DGP)

A data generating process (DGP) is a statistical model that describes how data is generated. It is a theoretical construct that helps us understand the underlying patterns and relationships in the data. In practice, we often do not know the true DGP, and we must rely on assumptions and approximations to make predictions.

Examples of DGPs include linear regression models, time series models, and stochastic processes. Understanding the DGP is crucial for selecting appropriate forecasting techniques and for interpreting the results of forecasts.

1.1 True DGP vs. Models

True DGP refers to the actual process that generates the data, while models are simplified representations of the DGP that we use for forecasting. Models may not capture all the complexities of the true DGP, but they can still provide useful insights and predictions. It is important to evaluate the performance of models against the true DGP to ensure their accuracy and reliability.

1.2 Illustration 1: DGP and Models - Study Hours and Exam Scores

Suppose we observe the following data for 10 students. Each student reports the number of hours they studied before an examination, and we observe their actual examination score.

Student	Study Hours	Actual Score
1	1.5	43
2	2.0	47
3	3.5	56
4	4.0	59
5	5.5	68
6	6.0	70

Student	Study Hours	Actual Score
7	7.5	78
8	8.0	81
9	9.5	87
10	10.0	89

At this stage, **we make no assumptions about how the data were generated.**

We simply observe the relationship between study hours and actual scores.

1.2.1 Data Display and Scatter Plot

Coding Session

```
#| echo: true #| output: true #| label: lst-dgp-1 #| lst-cap: "Data Display and Scatter Plot"
import pandas as pd import matplotlib.pyplot as plt
```

2 Create the dataset

```
df = pd.DataFrame({ "Student": range(1, 11), "Study_Hours": [1.5, 2.0, 3.5, 4.0, 5.5, 6.0, 7.5, 8.0, 9.5, 10.0], "Actual_Score": [43, 47, 56, 59, 68, 70, 78, 81, 87, 89] })
```

3 Display the data

```
print(df)
```

4 Create scatter plot

```
plt.figure(figsize=(8, 4.5))
plt.scatter( df["Study_Hours"], df["Actual_Score"], s=80 )
plt.xlabel("Study Hours", fontsize=11) plt.ylabel("Actual Score", fontsize=11) plt.title( "Relationship
Between Study Hours and Exam Scores", fontsize=13, pad=15 )
plt.grid(alpha=0.2) plt.tight_layout() plt.show()
```